

MGT-455R: Customer Analytics and (Generative) AI SP26

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DESCRIPTION

The course is very much hands-on, uses real-world data, and will equip you with tools and conceptual understanding needed leverage customer analytics to produce immediate value on the job. We will have 3 in-person sessions, online instructional materials (videos and readings), individual and group assignments, and exercises.

OBJECTIVES

Many firms possess extensive data on customer behavior, yet few have the expertise to translate this information into effective actions. Recent developments have made these capabilities even more valuable: The deprecation of tracking data, Generative AI, and the technical and economic feasibility of moving beyond traditional segmentation to individual-level customer connections at scale.

Internal Data as Strategic Asset

Privacy regulations such as GDPR, iOS restrictions, ad-blockers, and the advent of GenAI browsers are introducing huge changes to the data landscape. Organizations can no longer rely on external tracking to understand customer behavior. This makes internal data and the analysis of such to not merely useful but critical to support data-driven decision making. Firms that fail to fully leverage their own customer data operate at a structural disadvantage.

Generative AI, Personalization, and Individual-Level Analytics

Generative AI is democratizing the ability to conduct both basic and advanced analytics. Taking full advantage of these tools, however, requires deeper understanding of how data are generated, how to evaluate and apply output from machine learning models, and how to formulate questions that yield appropriate and valid conclusions. Practically, generative AI enables organizations to act on analytical insights at scale and lower cost—customer analytics can now inform personalized content generation, service interactions, and product recommendations at the individual level. Yet the quality of AI-generated personalization depends predominantly on the quality of the underlying customer-level analysis. Traditional marketing segmentation groups customers into broad categories for standardized products, services, and messaging. Customer analytics now makes individual-level analysis economically viable in both B2C and B2B contexts. Understanding individual customer preferences, lifetime value trajectories, and behavioral patterns allows organizations to deploy AI tools effectively. Without this analytical foundation, personalization efforts produce noise rather than value.

TOOLS

- Radiant is an (open-source) tool for business analytics that runs in a browser on Windows, Mac, and Linux. For additional information see <https://radiant-rstats.github.io/docs>

- Generative-AI tools to conduct data analysis are in active development and will be made available during the first class session. More generally, we will leverage GenAI extensively for learning, analytics, and guidance
- No programming experience is required for this class. That said, if you are interested learning more about coding in Python, you can get 6-months of free access by joining the Customer Analytics Group on DataCamp

MATERIALS

For this class you will work through a mix of videos, articles, assignments, and cases; no textbook is required. Lecture notes and additional materials will be available on the course website. Please refer to the appropriate page for the weekly modules. Generative AI will play a significant role in this class. I encourage you to integrate Generative AI into your personal and business activities to gain experience with this powerful and disruptive technology.

SCHEDULE

Module 01: Introduction to Customer Analytics (in-person)

Welcome to MGT455R: Customer Analytics and (Generative) AI. This week we will cover the fundamentals of Customer Analytics and discuss the evolving impact of GenAI on analytics and the future of work

Learning Objectives:

1. Understand the impact of the “information explosion” on customer relationships
2. Illustrate the connection between Business Analytics, Machine Learning, and (Generative) AI
3. Explain the importance of customer centricity for customer analytics
4. Discuss how Generative AI is changing customer analytics, and more broadly, how (data) work gets done
5. Emphasize the importance prescriptive analytics in data-driven decision making

1.1 Customer Analytics Overview

Objectives:

- Review and discuss key elements from the Course Introduction and Course Logistics videos
- Emphasize the importance of developing mutually beneficial relationships between organizations and individual customers
- Illustrate the importance of customer centricity for customer analytics

1.2 Basic Customer Analytics and Generative AI

Objectives:

- Refresh and enhance analytics skills
- Introduce GenAI tools for Customer Analytics

1.3 Targeting Offers Using Test & Rollout Campaigns

Objectives:

- Illustrate the benefits of customer targeting
- Learn how to implement a test and roll campaign

Module 02: Leveraging Generative AI for Analytics

This session refreshes core business analytics concepts and connects them to customer analytics workflows. We review how to read regression output, generate predictions, and interpret what the model is actually telling you. The emphasis is on translating output into decisions and how we can best leverage GenAI tools to support this process.

We then continue our discussion on logistic regression as a foundational model for predicting customer behavior and show how to score customers for targeting. Finally, we apply these ideas in a test-and-rollout setting to illustrate why targeting can outperform blanket messaging, and how experimental data supports practical rollout decisions.

Learning Objectives:

By the end of this week, you will be able to:

1. Interpret regression output and generate predictions
2. Understand Permutation Importance Plots (PIP) and Partial Dependence Plots (PDP)
3. Build and interpret logistic regression models
4. Score customers using logistic regression
5. Implement test and rollout campaigns

2.1 Group Formation

Important: Form groups of 3 (max) for upcoming group assignments

To join a group:

1. Click on “People” in Canvas left navigation
2. Click on the “Groups” tab
3. Find a group with space available
4. Click “Join” to add yourself

If you’re not in a group after the due date, you’ll be randomly assigned.

2.2 Business Analytics Review

Objectives:

- Refresh your memory of key business analytics concepts
- Review how to interpret regression output and generate predictions

2.3 Predicting Response with Logistic Regression

Objectives:

- Introduce a foundational model to predict customer choice: Logistic regression
- Understand how to score customers using logistic regression

2.4 Targeting Using Test & Rollout Campaigns

Objectives:

- Illustrate the benefits of customer targeting
- Learn how to implement a test and roll campaign
- Complete the basic customer analytics assignment. This assignment is about *getting to know* a customer database by conducting descriptive analytics using Radiant. The assignment is linked under “Cases” in the navigation bar.

3 Module 03: Targeting and Performance Evaluation

This session centers on the Tuango case and uses logistic regression to score customers for action. We focus on what the model implies for targeting decisions and how to connect the outputs to business outcomes. We then move to model evaluation using metrics such as lift, gains, and AUC.

Learning Objectives:

By the end of this week, you will be able to:

1. Design effective mobile app messaging strategies
2. Apply logistic regression to real-world customer targeting problems
3. Calculate ROI for targeting campaigns
4. Evaluate model performance using lifts and gains charts

3.1 Case Analysis: Tuango: Targeting Mobile App Messages (individual assignment)

Tuango is a daily deals platform that needs to optimize its mobile push messaging strategy. With limited messaging opportunities and customer attention, the company must carefully target which customers receive which offers.

Objectives:

- Understand how to score customers using logistic regression
- Design a targeting strategy that maximizes ROI

Reading:

The Backward Market Research article is excellent but unfortunately no longer available for free. Instead, read the Beginning at the End page and the worksheet

Preparation (individual assignment due):

- Complete the Tuango individual assignment. The assignment is linked under “Cases” in the navigation bar

3.2 Estimating and Evaluating Predictive Models**Objectives:**

- Learn to evaluate model performance comprehensively
- Understand lift and gains charts
- Compare different model evaluation metrics

4 Module 04: Developing Customers

The solution video for the Tuango case will be released after the due date for case and a set of follow up questions will be posted on the case page for you to complete. We then move to model evaluation and decision-focused metrics such as lift, gains, and AUC, and discuss how these measures guide targeting thresholds. This is a follow up on the reading on Estimating and Evaluating Predictive Models from Module 03.

Learning Objectives:

By the end of this week, you will be able to:

1. Deepen understanding of effective mobile app messaging strategies
2. Understand the relative complexity and value of Descriptive, Predictive, and Prescriptive Analytics
3. Extend your understanding of model performance evaluation from a business perspective
4. Understand what interaction effects are and their essential role in effective customer-level marketing

Plan ahead:

- Now that you have completed the Tuango case and we reviewed the solution, I suggest you start working on the “Intuit Quickbooks Upgrade” case with your group as soon as possible. Since this case is, by design, less structured than the preceding assignments make sure to leave yourselves enough time to think through the different challenges and develop a strong solution. The case is linked under “Cases” in the navigation bar
- It can be very useful to join the worksession on Monday’s (6-8pm) on zoom with your group to get direct support as you work through the case together.

5 Module 05: Targeting for Intuit Quickbooks**5.1 Intuit Quickbooks Upgrade (group assignment)**

Complete the Intuit Case with your group. The case is linked under “Cases” in the navigation.

5.2 Using Machine Learning for Customer Analytics**Objectives:**

- Introduce neural networks and their application in Customer Analytics

- Discuss how insights from machine learning models, such as Neural Networks, can be used to augment *Analyst Driven Models*, such as Linear and Logistic Regression

6 Module 06: Intuit Quickbooks Redux

6.1 Intuit Quickbooks Upgrade: Redux (optional group case for extra credit)

You have the option to resubmit the intuit case with new and improved analytics. The case is linked under “Cases” in the navigation bar

6.2 Using Machine Learning for Customer Analytics (continued)

Objectives:

- Continue overview of machine learning models and their application in Customer Analytics
- Demonstrate how insights from machine learning models can be used to augment *Analyst Driven Models*, such as Linear and Logistic Regression

6.3 Peer Evaluation Form

The peer evaluation form will be available at <https://rsm-django-02.ucsd.edu/peerevals/>

7 Module 07: Measuring Uplift for Creative Gaming

7.1 Propensity-to-Buy vs Uplift Modeling

Objectives:

- Learn the difference between propensity and uplift
- Evaluate the pros and cons of propensity-to-by versus an uplift approach to targeting
- Evaluate the quality of an uplift model

7.2 Case Analysis: Creative Gaming: Uplift Modeling (group assignment)

- Complete the Creative Gaming case using uplift modeling. The group case is linked under “Cases” in the navigation bar
- The data for the uplift demo is available for download. Make sure to read the creativegaming.pdf document included in the case zip-folder for more information on the data and the case.

Objectives:

- Practice targeting customers based on uplift
- Evaluate the quality of an uplift model
- Evaluate the pros and cons of propensity-to-by versus an uplift approach to targeting

Preparation (Group Assignment Due):

- Prepare “Creative Gaming: Uplift Modeling” for class discussion and as group assignment

Module 08: Personalization for Pentathlon

8.1 Hyper-Personalization: Next-Product-to-Buy Models

Objectives:

- Understand the benefits of using customer purchase data for customer development
- Understand how to cross-sell and up-sell
- Illustrate the benefits of product and message tailoring

Readings:

- Stitch Fix CTO: Hyper-personalization and Creating a Store for One

8.2 Churn and CLV

Objectives:

- Recognize that customer retention is an important problem in many industries
- Learn how analytical methods can be used to reduce churn
- Understand how Customer Lifetime Value (CLV) calculations can guide business decisions

- Calculate Customer Lifetime Value (CLV)

Audio:

- Canceling an AOL account, or, how **not** to manage customer churn

Preparation:

- Review the calculations in the xlsx file for *AWS* and for *NotFlix*
- Predicting Churn and Churn Management
- Customer Lifetime Value (CLV) Calculations

9 Module 09: Causality and Pitfalls (in-person)

9.1 Case Analysis: Pentathlon: Next Product to Buy Modeling

Objectives:

- Understand the benefits of using customer purchase data for customer development
- Demonstrate the benefits of product and message tailoring

Preparation (Group Assignment Due):

- Prepare “Pentathlon: Next Product to Buy Modeling” for class discussion and as group assignment

9.2 The Causality Checklist

Objectives:

- Avoid interpreting analysis results incorrectly
- Evaluate if analysis results (can) have a causal interpretation

Required Reading:

Read Pentathlon: Promotional E-mail Frequency

9.3 Course Wrapup

Objectives:

- Avoid common customer analytics implementation pitfalls
- Review and synthesize insight from “MGT455R: Customer Analytics and (Generative) AI”

10 Module 10: Choose your own Adventure

This is the last week of the quarter for this class. You have **one** final deliverable that can be either individual or with your group. You have three options to choose from, and you can select the one that is most interesting and relevant to you and your group. The three options are:

1. S-mobile: Predicting Customer Churn (group case)
2. FiTech: Data Driven Credit Card Design (group case)
3. Customer Analytics and GenAI Integration plan (individual exercise)

10.1 Option 1: S-mobile: Predicting Customer Churn

Objectives:

- Understand how to predict customer attrition (i.e., churn)
- Learn how to use insights from a churn model to develop an incentive plan that can entice would-be churners to remain customers
- Understand how to translate predictions into actions

Preparation (Group Assignment):

- Prepare “S-Mobile: Predicting Customer Churn” for class discussion and as group assignment.

10.2 Option 2: FiTech: Data Driven Credit Card Design

Objectives:

- Illustrate the benefits of product tailoring
- Evaluate the relative costs and benefits of testing (RCT) and using historical data

Preparation (Group Assignment):

- Prepare “FiTech: Data Driven Credit Card Design” for class discussion and as a group assignment.

10.3 Option 3: Customer Analytics and GenAI Integration Plan

Create a concise plan to integrate what you have learned about Customer Analytics (and Generative AI) into your organization. What elements would be most (least) useful in your organization and why? Make sure to discuss implementation challenges and how they could be addressed. Aim for no more than 4 pages of single-space text (not including appendices and exhibits).

10.4 Peer Evaluation Form

The peer evaluation form will be available at <https://rsm-django-02.ucsd.edu/peerevals/>

10.5 Course Evaluations

Don't forget to complete the course evaluation for MGT455R: Customer Analytics and Generative AI: <https://academicaffairs.ucsd.edu/Modules/Evals/>

GRADING

Grades for this course will be based on the following components:

- Individual assignments (20%)
 - Selecting Prospects (5%)
 - Basic Customer Analytics (5%)
 - Tuango: Targeting Mobile App Messages (10%)
- Group assignments (40%)
 - Intuit Quickbooks Upgrade (10%)
 - Creative Gaming: Uplift Modeling (10%)
 - Pentathlon: Next Product to Buy Modeling (10%)
 - Choice of individual or group case (10%)
- Class participation (40%)
 - The rubric that will be used to evaluate class participation is shown below
 - Completed TODOs on the course website, excluding assignments and cases, will be counted towards the class participation grade
 - 20-points will be deducted if peer evaluations are not submitted

Score	Description
3 (Excellent)	Student actively contributes to class discussions with insightful comments, demonstrating thorough preparation and critical thinking. Consistently applies course concepts, engages respectfully with peers, and significantly enhances the learning environment.
2 (Satisfactory)	Student regularly participates with relevant ideas, showing adequate preparation. Often applies course material and engages positively with classmates. Maintains a respectful presence and contributes to productive discussions.
1 (Needs Improvement)	Student occasionally contributes but comments may lack depth or relevance. Shows inconsistent preparation and limited application of course concepts. Minimal engagement with peers. May sometimes be inattentive. Students attending and answering all iClicker questions will earn 1 point for class participation.
0 (Unsatisfactory)	Student rarely or never participates, shows little to no preparation, and does not engage in class discussions. May be disruptive or disrespectful. Includes absent students. Includes students that attend class but did not participate in iClicker questions.

Assignments submitted after the due date cannot be accepted. All regrade requests should be submitted no more than two weeks after the grade was posted. TAs will not consider in-person regrade requests.

COURSE POLICIES

Generative Artificial Intelligence (AI) policy

The use of GenAI tools is encouraged in this class. These tools can help to streamline and augment your work. Use them with the following caveats and conditions:

- GenAI-tools should enhance your work, not replace it. For instance, large language models can improve the quality of your writing, identify new sources, provide high-level summaries of complex ideas or terms, generate code, etc. etc. You should not use GenAI tools to generate content that you simply put your name on. This constitutes an academic integrity violation, much like taking another individual's work and misrepresenting it as your own.
- We expect that you understand and can answer detailed questions about the work you submit as your own.
- Cold-calling will be used extensively in class and you may be asked to explain the work you submitted.
- If you use GenAI for an assignment, be transparent about how it was used in a footnote or citation (e.g., a markdown file with the chat log).
- Large language Models (LLMs) have documented issues with hallucination and inaccuracy. You are responsible for all submitted content and will be graded accordingly.

Class Etiquette and Punctuality

- Laptops may not be used in class unless it is required for an in-class exercise. It is distracting to classmates to sit next to or behind someone while they browse the web, send emails, etc. However, tablets may be used to take notes.
- Phones can be used only to answer polling questions.

- Class will start on time. It is distracting to classmates and to the instructor when someone arrives late, has to get to their seat, and settle in while others are trying to engage with the class material. If you are absent or late more than once, your class participation grade will be affected.

Attendance

Learning to present your work and to evaluate and respond to the work conducted by others is an important part of this class. If you miss class, you will miss this learning opportunity, and there is not an easy way to “make it up.” As a result, you should make every effort to attend class. If you miss class or are late more than once, it will lower your class participation grade.

Note that UCSD provides exceptions for religious holidays, funeral attendance, and student or dependent hospitalization. If you must miss a class, please let us know in advance. You should do the readings, watch videos, prepare, turn in the assignments on time (late assignments will not be accepted), and arrange to get notes from a fellow student about what you missed in class.

Work Sessions

We will have optional work sessions each week, accessible through zoom. You can join any time between 5pm and 8pm. Although these sessions are optional I highly recommend you come to ask questions, discuss class material, work on assignments and cases, either alone or with your group, etc.

Support

For questions about the class, please post to the Piazza site linked below.

<https://piazza.com/ucsd/winter2026/mgt455r>

- School: UCSD
- Term: Winter 2026
- Class: mgt455r

When you post a question to Piazza, please make the post “public” whenever possible. It is fine if you prefer to be anonymous to your classmates. That is up to you. However, we will get the greatest value out of using Piazza if (most) posts are public.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.